EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

Spring 2026

EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

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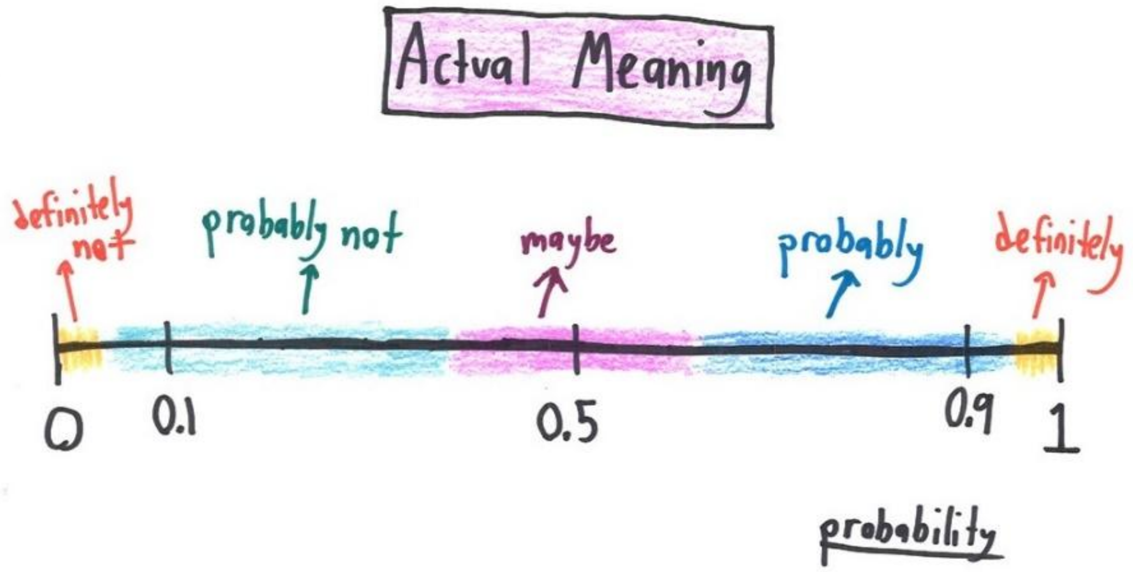
Week 1 - Lecture No 02 – January 14, 2026



Announcements!

1. Lecture No 01 posted.
2. **Quiz No 01 – January 21, 2026**
3. **Office hours – Tuesday & Thursday 12:00 to 1:00 PM @ Room C-211 (or through meeting request)**
4. **Supplemental Reading on Set Theory uploaded**

Agenda for today

1. Unit 1 – Probability models and axioms

Unit 1: Probability models and axioms – Lecture outline

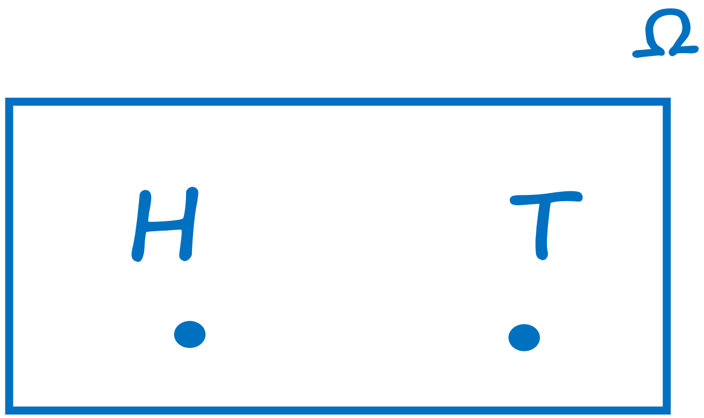
- Sample space
- Probability laws
  - Axioms
  - Some properties
- Examples
  - Discrete
  - Continuous
- Interpretations of probabilities

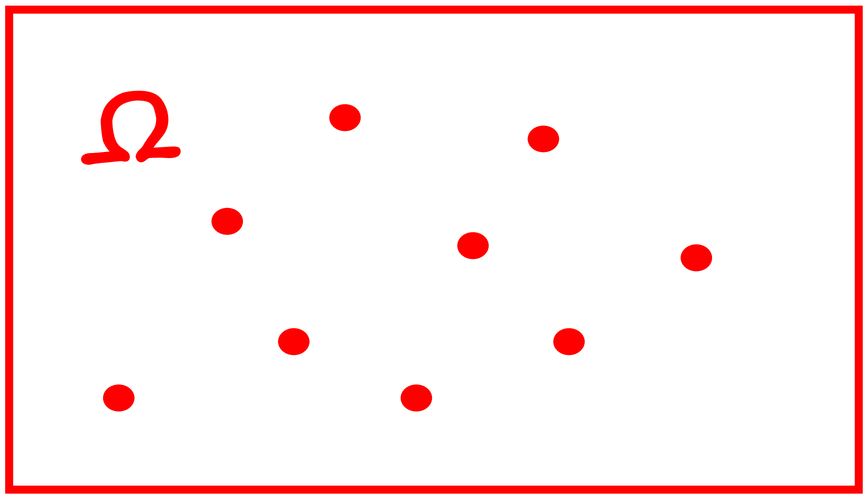
Sample space

- Two steps
    - Describe ***possible outcomes***
    - Describe ***beliefs about likelihood*** of possible outcomes

Sample space

- List (set) of possible outcomes, Ω
- List must be:
  - Mutually exclusive
  - Collectively exhaustive
  - At the "right granularity"



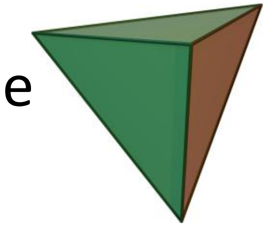


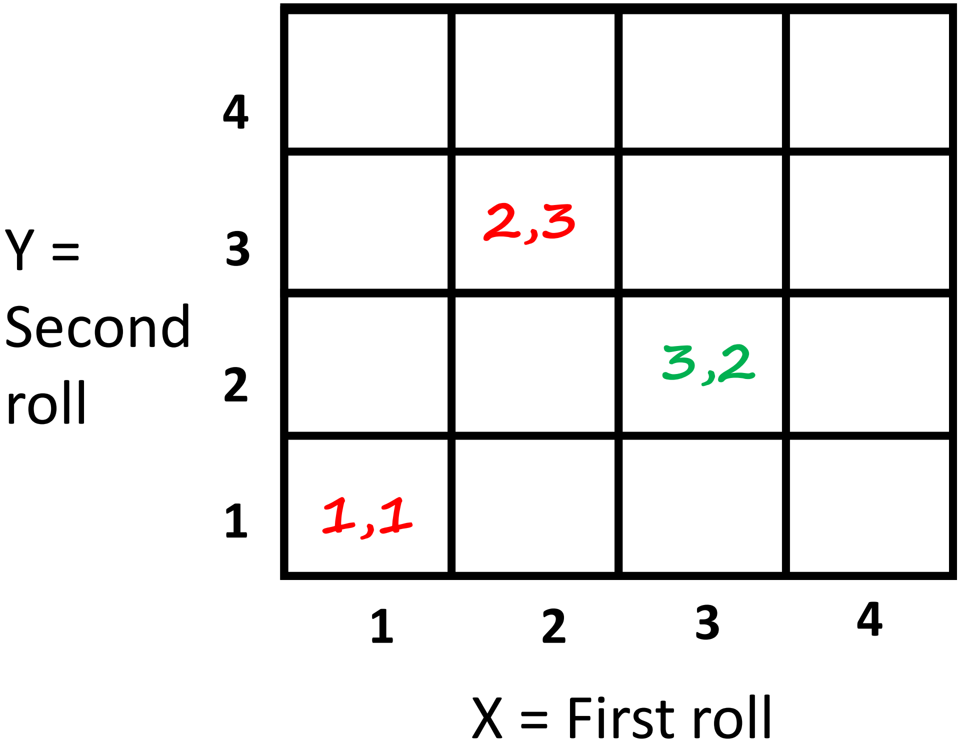
Ω

- H and rains
- H and no rain
- T and rains
- T and no rain

Sample space: discrete/finite example

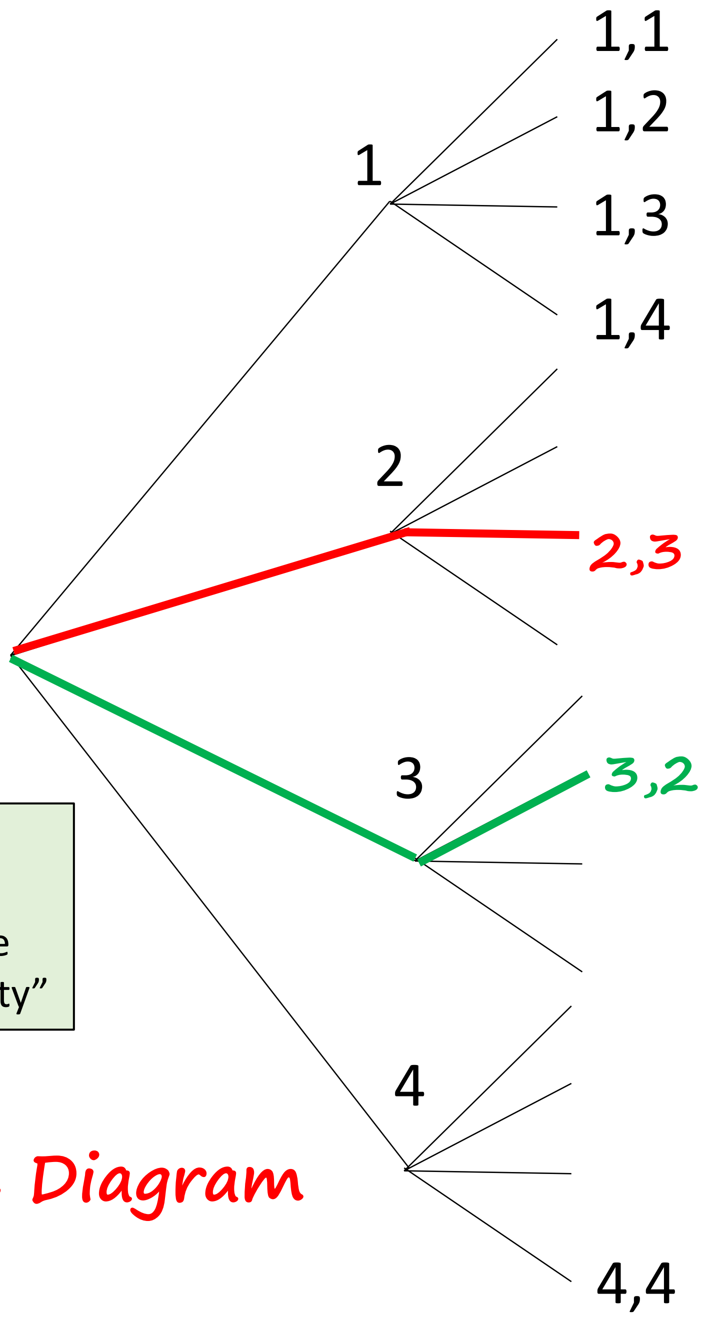
- Two rolls of a tetrahedral die





- List must be:
  - Mutually exclusive
  - Collectively exhaustive
  - At the “right granularity”

Sequential description



Tree Diagram

Exercise No 01!

Suppose you roll two fair, six-sided dice, one of which is red and the other of which is green. Define the following two random variables X & Y:

Example

X =
Y =

X = The number shown on the red die

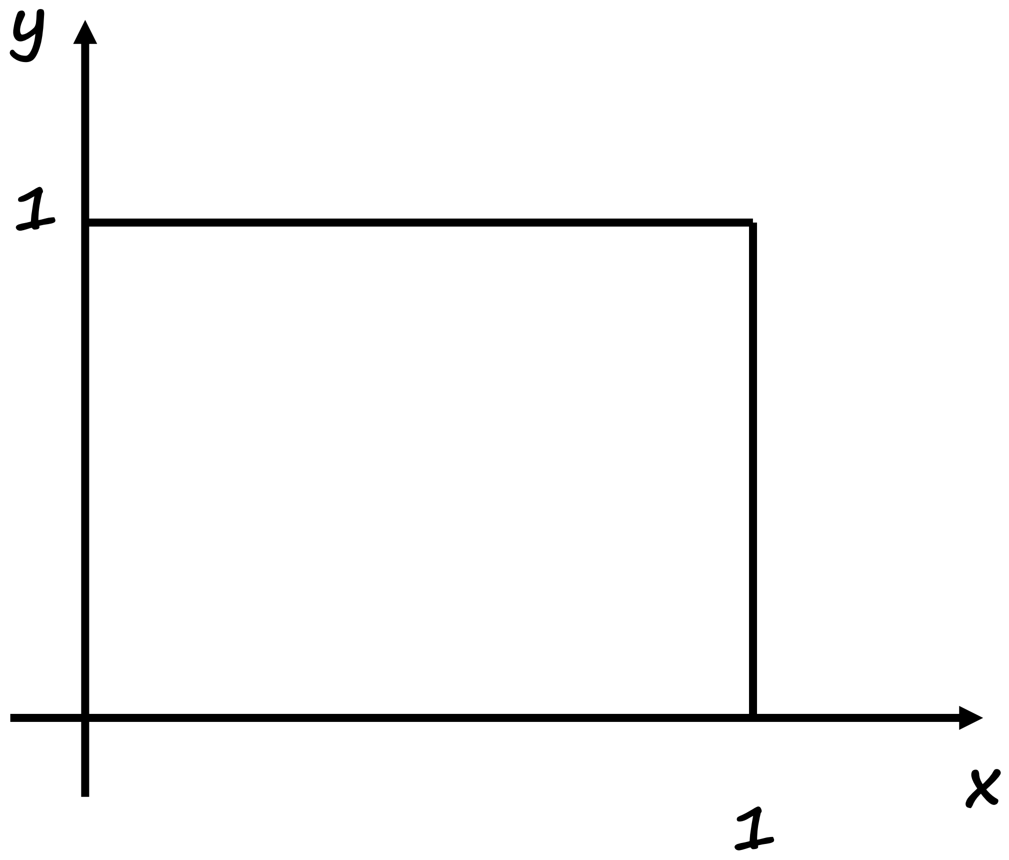
$$Y = \begin{cases} 0 \text{ if the two dice show the same number} \\ 1 \text{ if the number on the green die is bigger than the number on the red die} \\ 2 \text{ if the number on the red die is bigger than the number on the green die} \end{cases}$$

- Write down a table capturing the Sample Space & joint probabilities for X and Y .

	$X =$ 1	2	3	4	5	6
$Y =$ 0						
1						
2						

Sample space: continuous example

- (x,y) such that $0 \le x,y \le 1$



Probabilistic Model

- Two steps
  - Describe possible outcomes – Sample Space
  - **Describe beliefs about likelihood of outcomes -**

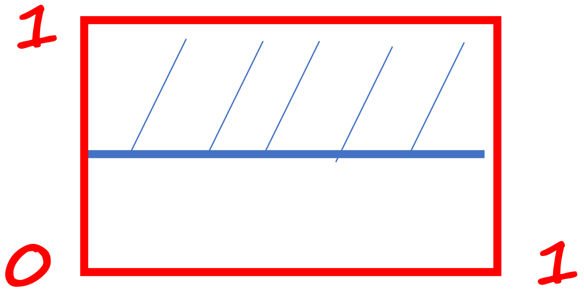
Probability axioms

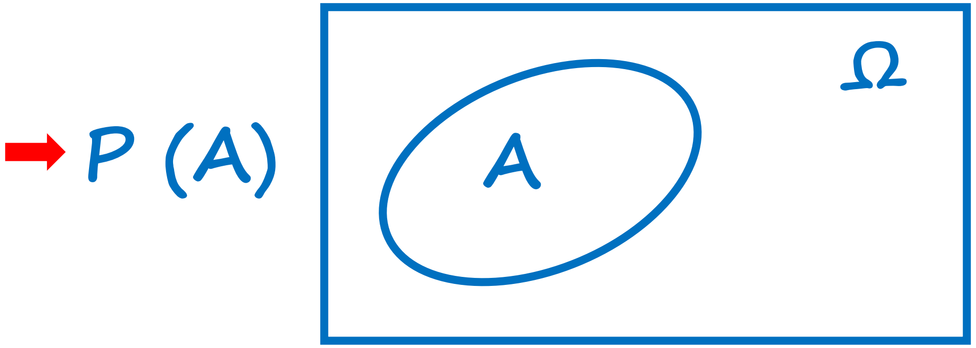
- Event: a subset of the sample space
  - Probability is assigned to events

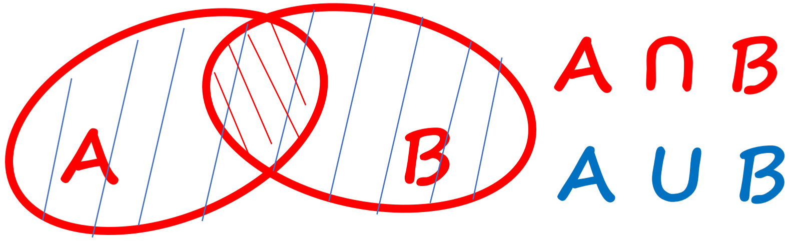
- Axioms:
  - Nonnegativity: $\mathbf{P}(A) \geq 0$
  - Normalization: $\mathbf{P}(\Omega) = 1$
  - (Finite) additivity: (to be strengthened later)

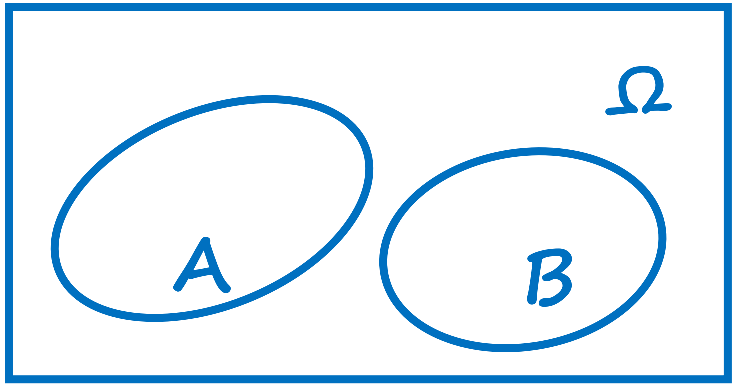
    If $A \cap B = \emptyset$, then $\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$

    empty set









Some simple consequences of the axioms

Axioms

$\mathrm{P}\,(A) \geq 0$

$\mathrm{P}\,(\Omega) = 1$

For disjoint events:

$\mathrm{P}\,(A \cup B) = \mathrm{P}\,(A) + \mathrm{P}\,(B)$

Consequences

$\mathrm{P}\,(A) \leq 1$

$\mathrm{P}\,(\emptyset) = 0$

$\mathrm{P}\,(A) + \mathrm{P}\,(A^c) = 1$

$\mathrm{P}\,(A \cup B \cup C) = \mathrm{P}\,(A) + \mathrm{P}\,(B) + \mathrm{P}\,(C)$

and similarly for *k* disjoint events

$$\mathrm{P}\,(\{s_1, s_2, \ldots, s_k\}) = \mathrm{P}(\{s_1\}) + \ldots + \mathrm{P}(\{s_k\})$$
$$= \mathrm{P}(s_1) + \ldots + \mathrm{P}(s_k)$$

Some simple consequences of the axioms

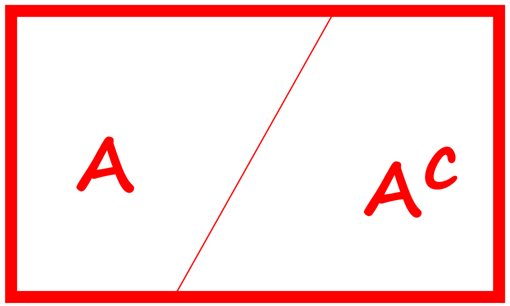
Axioms

a) $P(A) \geq 0$

b) $P(\Omega) = 1$

For disjoint events:

c) $P(A \cup B) = P(A) + P(B)$



$A \cup A^c = \Omega$

$A \cap A^c = \emptyset$

$\Rightarrow 1 \overset{(b)}{=} P(\Omega) = P(A \cup A^c)$

$\Rightarrow \overset{(c)}{=} P(A) + P(A^c)$

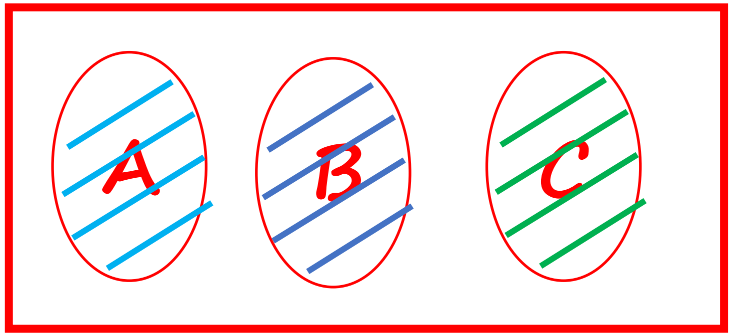
$\Rightarrow P(A) = 1 - P(A^c) \overset{(a)}{\leq} 1$

$\Rightarrow 1 = P(\Omega) + P(\Omega^c)$

$\Rightarrow 1 = 1 + P(\emptyset) \Rightarrow P(\emptyset) = 0$

Some simple consequences of the axioms

- A, B, C disjoint: P (A U B U C) = P (A) + P (B) + P (C)

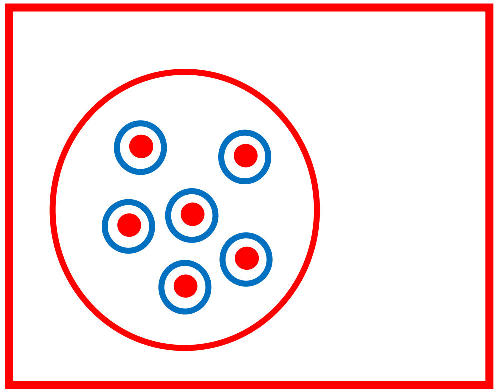


$P(A \cup B \cup C) = P((A \cup B) \cup C) = P(A \cup B) + P(C)$

$\Rightarrow \quad = P(A) + P(B) + P(C)$

$\Rightarrow$ If $A_1, \dots A_k$ disjoint $\Rightarrow P\left(A_1 \cup \dots \cup A_k\right) = \sum_{i=1}^{k} P(A_i)$

- P $(\{s_1, s_2, \dots, s_k\})$ = $P(\{s_1\} \cup \{s_2\} \cup \dots \cup \{s_k\})$

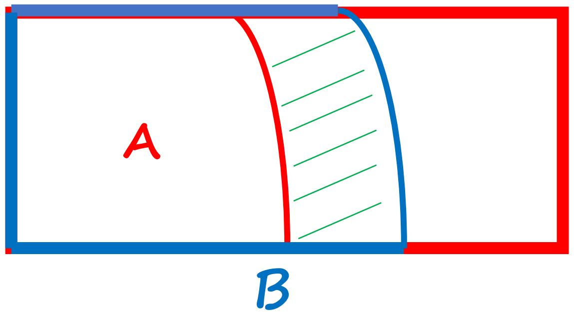


$= P(\{s_1\} + \{s_2\} + \dots + \{s_k\})$

$= P(s_1) + P(s_2) + \dots + P(s_k)$

More consequences of the axioms

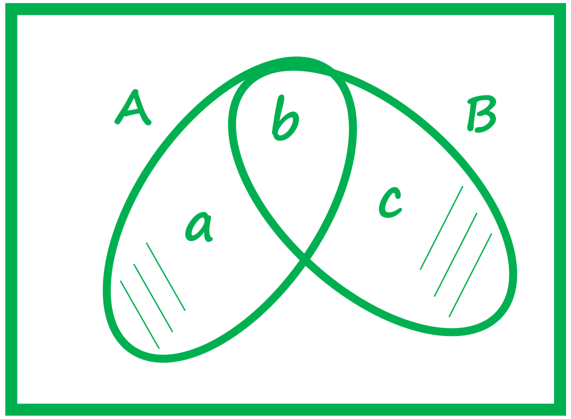
- If $A \subset B$, then $\mathrm{P}(A) \leq \mathrm{P}(B)$



$\Rightarrow B = A \cup (B \cap A^c)$

$P(B) = P(A) + \underline{P(B \cap A^c)} \geq P(A)$

- $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \overbrace{\mathrm{P}(A \cap B)}^{\geq 0}$



$a = P(A \cap B^c) \quad b = P(A \cap B) \quad c = P(B \cap A^c)$

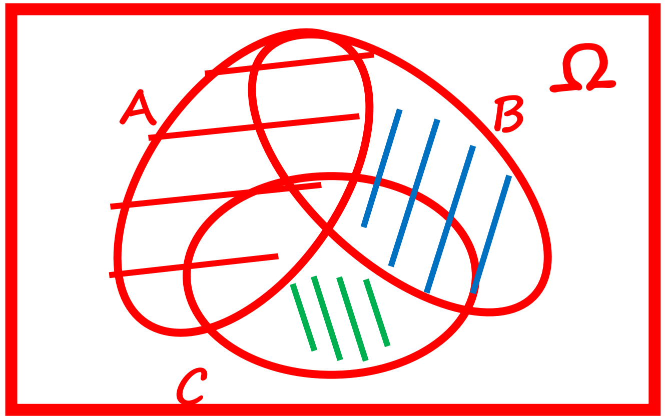
$P(A \cup B) = a + b + c$

$$P(A) + P(B) - P(A \cap B) = (a + b) + (b + c) - b$$
$$= a + b + c$$

- $\mathrm{P}(A \cup B) \leq \mathrm{P}(A) + \mathrm{P}(B)$ union bound

More consequences of the axioms

- $P(A \cup B \cup C) = P(A) + P(A^C \cap B) + P(A^C \cap B^C \cap C)$

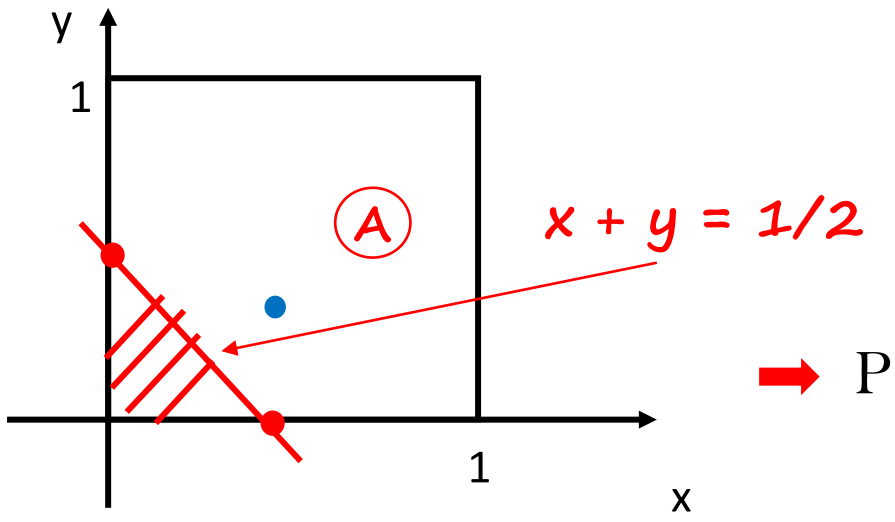


$\Rightarrow P(A \cup B \cup C) =$

$\Rightarrow \quad = P(A \cup (B \cap A^C) \cup (C \cap A^C \cap B^C))$

Probability calculation: continuous example

- (x, y) such that $0 \le x, y \le 1$
- **Uniform** probability law: Probability = Area



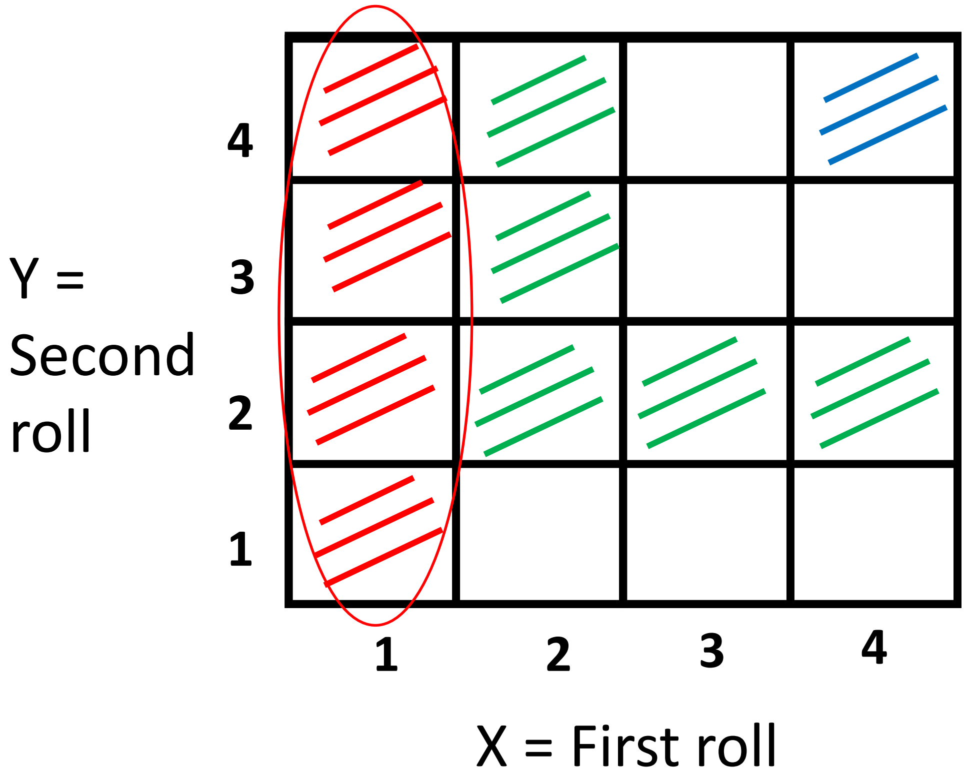
➡ $P(\{(x, y) \mid x + y \le 1/2\}) = \frac{1}{2} * \frac{1}{2} * \frac{1}{2} = \frac{1}{8}$

➡ $P(\{(0.5, 0.3)\}) = 0$

$$P(\{(x, y) \mid x + y = 1/2\}) = ?$$

Probability calculation: discrete/finite example

- Two rolls of a tetrahedral die



- Let every possible outcome have probability 1/16

- $\mathrm{P}\,(X = 1) = 4 * \frac{1}{16} = \frac{1}{4}$

Let $Z = \min\,(X, Y)$

$X=2,\ Y=3,\ \rightarrow Z=2$

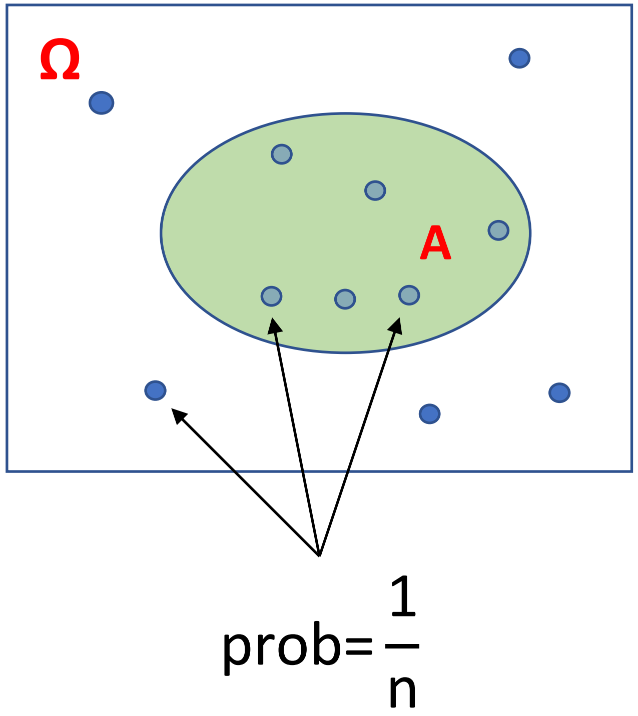
- $\mathrm{P}\,(Z = 4) = \frac{1}{16}$

- $\mathrm{P}\,(Z = 2) = 5 * \frac{1}{16}$

Discrete uniform law

finite

- Assume Ω consists of n equally likely elements
- Assume A consists of k elements



$$P(A) = k * \frac{1}{n}$$

Probability calculation steps (sequence of 4 steps)

1. Specify the sample space

2. Specify the probability law

3. Identify an event of interest

4. Calculate…

Probability calculation: discrete but infinite sample space

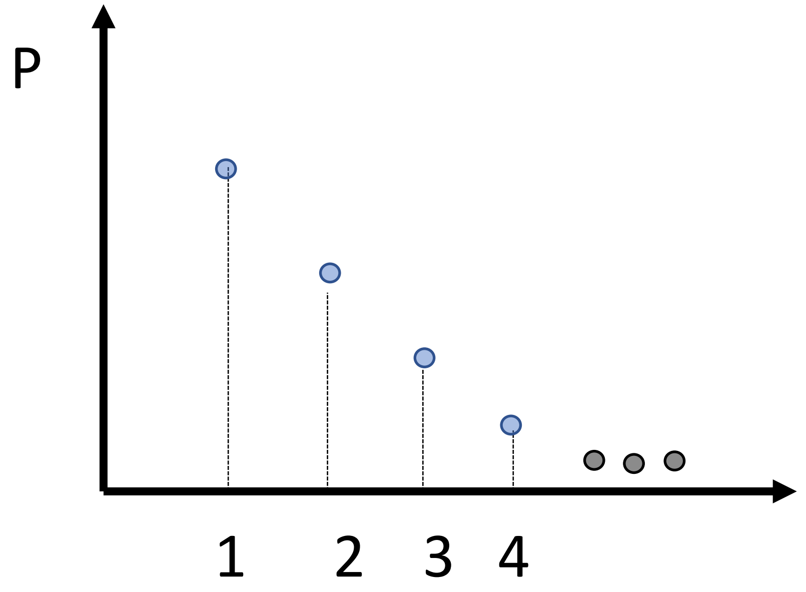
Specify the sample space

1. Sample space: {1, 2, . . . }

Specify the probability law

2. We are given $P(n) = \frac{1}{2^n}$, n=1, 2, . . .

$$\Rightarrow \sum_{n=1}^{\infty} \frac{1}{2^n} = \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2^n} = \frac{1}{2} * \frac{1}{1-\frac{1}{2}} = 1$$



Identify an event of interest

3. P (outcome is even) = $P(\{2, 4, 6, . . . \})$

Calculate the probability

$$= P(\{2\} \cup \{4\} \cup \{6\} \cup \ldots) = P(2) + P(4) + P(6) + \ldots$$

$$= \frac{1}{2^2} + \frac{1}{2^4} + \frac{1}{2^6} + \ldots = \frac{1}{4}\left(1 + \frac{1}{4} + \frac{1}{4^2} + \ldots\right) = \frac{1}{4} * \frac{1}{1-\frac{1}{4}} = \frac{1}{3}$$